



Answers to Statistics Tutorial Practice Questions

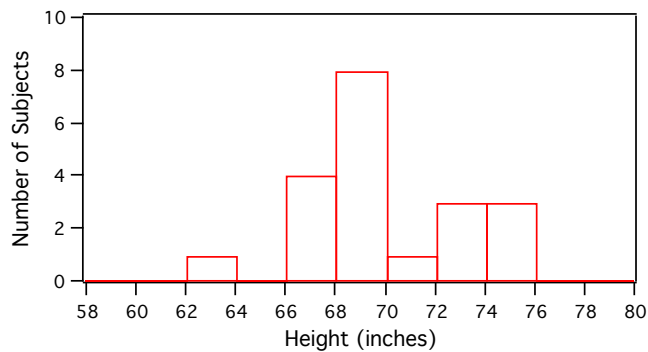
1) The mean is $(72.2 + 71 + 72.8 + 63.8 + 68.1 + 69.4 + 74.2 + 75.8 + 66.3 + 66.5 + 69.2 + 66.7 + 69.4 + 75.1 + 72.3 + 68.5 + 68.7 + 68 + 68.5 + 68.6) / 20 = 69.8$ inches

The standard deviation is $\sqrt{\frac{1}{19}[(72.2 - 69.8)^2 + (71 - 69.8)^2 + \dots]} = 3.1$ inches

2) We saw in question 1) that the standard deviation (S.D.) is 3.1 inches. From this, the standard error is

$$S.E. = \frac{S.D.}{\sqrt{20}} = \frac{3.1}{\sqrt{20}} = 0.7 \text{ inches}$$

3)



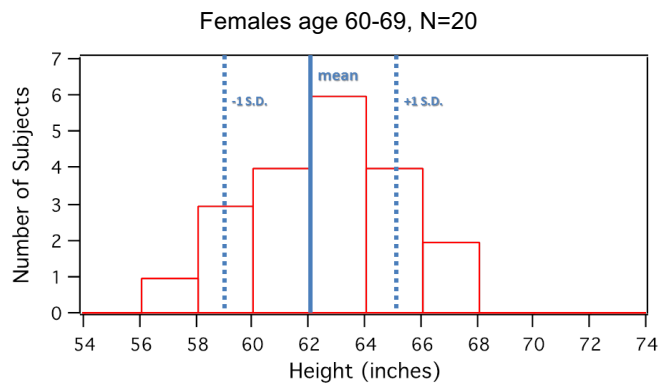
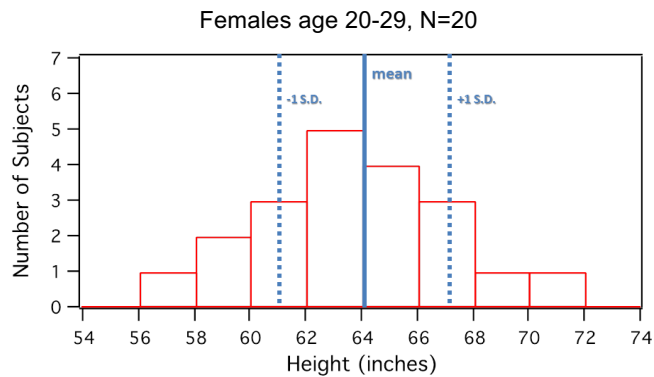
4) Choosing the mean ($\bar{X}$) such that it has the same number of bins with non-zero values on its two sides, and the standard deviation S.D. such that it excludes roughly $N/3=20/3=6.7$ subjects, we get (see also the graphs below)

$$\bar{X} \text{ (females age 20-29)} = 64 \text{ inches}$$

$$S.D. \text{ (females age 20-29)} = 3 \text{ inches}$$

$$\bar{X} \text{ (females age 60-69)} = 62 \text{ inches}$$

$$S.D. \text{ (females age 60-69)} = 3 \text{ inches}$$



5) The standard errors of the mean of the two samples above are

$$\text{S.E.}(\text{females age 20 to 29}) = \text{S.D.}(\text{females age 20 to 29}) / \sqrt{20} = 0.7 \text{ inches}$$

$$\text{S.E.}(\text{females age 60 to 69}) = \text{S.D.}(\text{females age 60 to 69}) / \sqrt{20} = 0.7 \text{ inches}$$

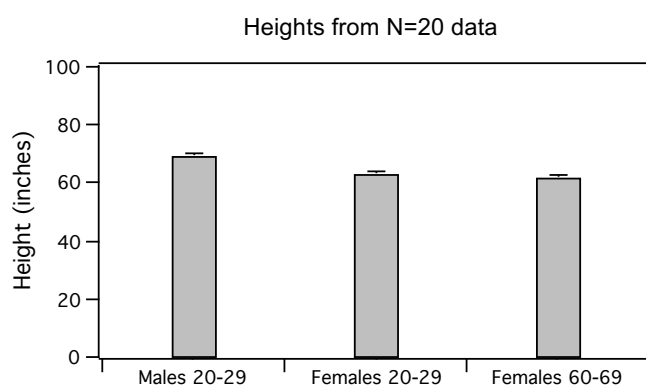
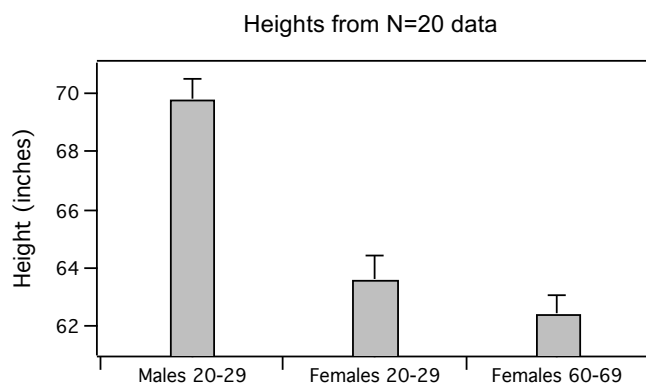
Estimates for the 95% confidence intervals (i.e., roughly the interval from $\bar{X} - 2\text{S.E.}$ to $\bar{X} + 2\text{S.E.}$) of the two groups are therefore

Females age 20-29 → from $64 - (2 \times 0.7) = 62.6$ inches to $64 + (2 \times 0.7) = 65.4$ inches

Females age 60-69 → from $62 - (2 \times 0.7) = 60.6$ inches to $62 + (2 \times 0.7) = 63.4$ inches

We see that the means of the two groups **do not** differ by a statistically significant amount since the 95% confidence intervals of the two groups overlap.

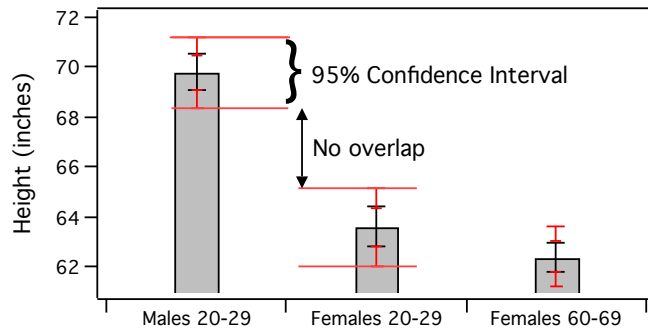
6)



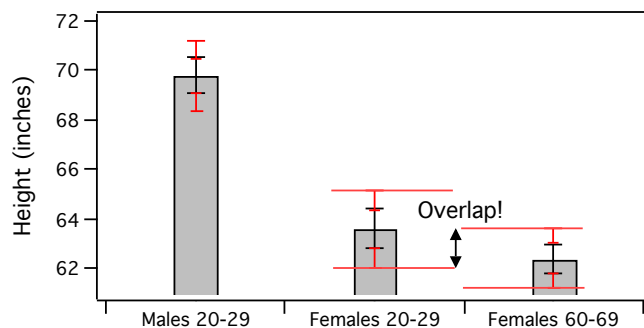
The heights of the bars represent the means, the error bars are 1 standard error. (The values here are plotted from the exact data, rather than the estimates. So they may vary slightly from your answer.)

Note that these two graphs plot the same data; the only difference between the two is the scaling of the y-axis. The first plot (the “zoomed-in” view) will prove most useful for answering question 7, because it allows us to examine the differences between the three means more closely. However, it may also be used deceptively, to overemphasize the differences. From a quick glance at the zoomed-in graph it appears that the value for “Males 20-29” is several times that for “Females 20-29”. The second graph, with the y-axis scaling starting at zero, gives a better impression of the overall comparison. Watch out for this trick, a visual dramatization of small differences that is common in news infographics.

7)



First, construct ~95% confidence intervals by drawing two standard errors up and down from the mean (shown in red). The ~95% confidence intervals for the mean heights of males in their 20s and females in their 20s do NOT overlap. So we can conclude the difference between the mean heights IS statistically significant.



The ~95% confidence intervals for the mean heights of females in their 20s and females in their 69s DO overlap. We conclude therefore that the difference in means is NOT statistically significant.

8) Choosing the mean ($\bar{X}$) such that it has the same number of bins with non-zero values on its two sides, the standard deviation S.D. such that it excludes roughly $N/3$ subjects, and their standard error S.E. such that it is equal to $S.D./\sqrt{N}$, we get (see also the graphs below)

$$\bar{X}(\text{males age 20-29}) = 70 \text{ inches}$$

$$\text{S.D.}(\text{males age 20-29}) = 4 \text{ inches}$$

$$\text{S.E.}(\text{males age 20-29}) = 4/\sqrt{808} = 0.14 \text{ inches}$$

$$\bar{X}(\text{females age 20-29}) = 64 \text{ inches}$$

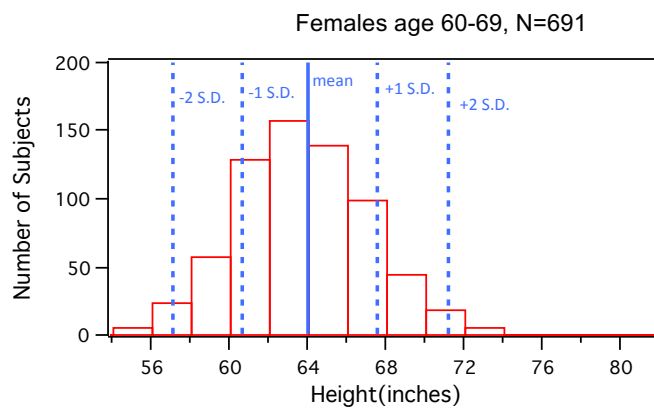
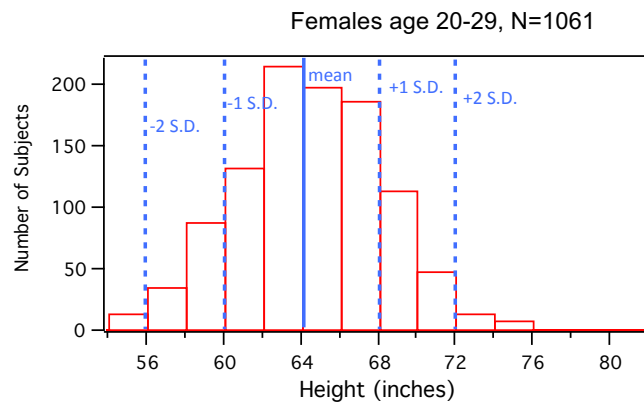
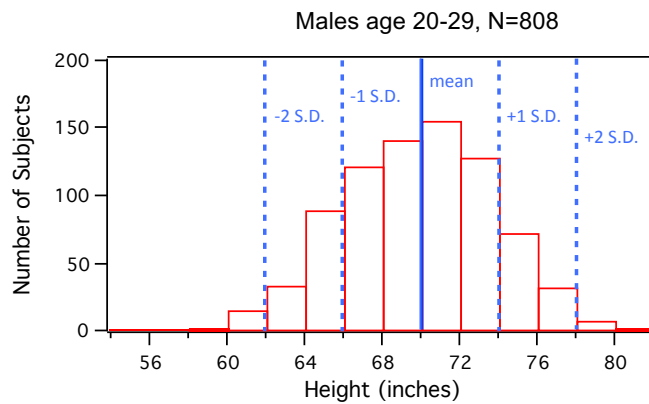
$$\text{S.D.}(\text{females age 20-29}) = 4 \text{ inches}$$

$$\text{S.E. (females age 20-29)} = 4 / \sqrt{1061} = 0.11 \text{ inches}$$

$$\bar{X} \text{ (females age 60-69)} = 64 \text{ inches}$$

$$\text{S.D. (females age 60-69)} = 3.5 \text{ inches}$$

$$\text{S.E. (females age 60-69)} = 3.5 / \sqrt{691} = 0.13 \text{ inches}$$

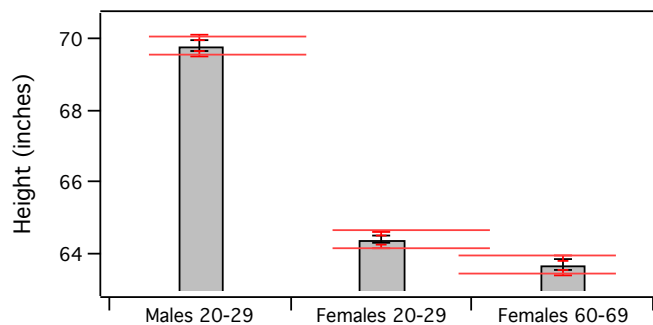


9)



Heights of bars represent means for each group, error bars are 1 standard error.

10)



First, we draw the interval from the mean minus two standard errors to the mean plus two standard errors to approximate the 95% confidence interval for each mean (shown in red). Since the 95% confidence intervals don't overlap for any of the populations, we can conclude that all three means have a statistically significant difference from each other.

11) The standard error is proportional to $1/\sqrt{N}$, so as the sample size increases, the standard error decreases. We can make the standard error as small as we like simply by choosing a large enough sample size.

With increasing sample size, the mean and standard deviation will also change as they become more precise representations of the true mean and standard deviation of the data. Unlike the standard error, the mean and standard deviation do not scale with sample size, so we can't predict whether they will increase or decrease, and we don't expect them to change much with increasing sample size.



Answers to Calculating with Units Tutorial Practice Questions

Part 1:

$$1) \frac{2 \cdot 7}{3 \cdot 5} = \frac{14}{15}$$

$$2) \frac{2}{3 \cdot 5} = \frac{2}{15}$$

$$3) \frac{2 \cdot 5}{3} = \frac{10}{3}$$

$$4) \frac{a}{b \cdot c}$$

$$5) \frac{d \cdot f}{e}$$

$$6) \frac{a \cdot d}{b \cdot c}$$

Part 2:

1)

$$5 \text{ days} = 5 \text{ days} \cdot \frac{24 \text{ hours}}{1 \text{ day}} = \frac{5 \text{ days}}{1} \cdot \frac{24 \text{ hours}}{1 \text{ day}} = 5 \cdot 24 \text{ hours} = 120 \text{ hours}$$

2)

$$1 \text{ day} = 1 \text{ day} \cdot \frac{24 \text{ hrs}}{1 \text{ day}} \cdot \frac{60 \text{ min}}{1 \text{ hr}} \cdot \frac{60 \text{ sec}}{1 \text{ min}} = 86400 \text{ sec} = 8.64 \times 10^4 \text{ seconds}$$

3)

$$2.5 \text{ hours} = 2.5 \text{ hrs} \cdot \frac{60 \text{ min}}{1 \text{ hr}} \cdot \frac{60 \text{ sec}}{1 \text{ min}} = 9000 \text{ sec} = 9 \times 10^3 \text{ seconds}$$

4)

$$10 \text{ min} = 10 \text{ min} \cdot \frac{1 \text{ hr}}{60 \text{ min}} \cdot \frac{1 \text{ day}}{24 \text{ hrs}} = 0.00694 \text{ days} = 6.94 \times 10^{-3} \text{ days}$$

5)

$$1 \text{ year} = 1 \text{ year} \cdot \frac{365 \text{ days}}{1 \text{ year}} \cdot \frac{24 \text{ hrs}}{1 \text{ day}} \cdot \frac{60 \text{ minutes}}{1 \text{ hour}} = 525,600 \text{ minutes}$$

Part 3:

1)

$$26.2 \text{ miles} = 26.2 \text{ mi} \cdot \frac{1.61 \text{ km}}{1 \text{ mi}} = \frac{26.2 \cancel{\text{mi}}}{1} \cdot \frac{1.61 \text{ km}}{1 \cancel{\text{mi}}} = 41.2 \text{ km}$$

2)

$$100 \text{ meters} = 100 \text{ m} \cdot \frac{1 \text{ ft}}{12 \text{ in}} \cdot \frac{1 \text{ in}}{2.54 \text{ cm}} \cdot \frac{100 \text{ cm}}{1 \text{ m}} = 328 \text{ feet}$$

3)

$$5000 \text{ miles} = 5000 \text{ mi} \cdot \frac{5280 \text{ ft}}{1 \text{ mi}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 316800000 \text{ in} = 3.17 \times 10^8 \text{ inches}$$

4)

$$\begin{aligned} 5000 \text{ miles} &= 5000 \text{ mi} \cdot \frac{1.61 \text{ km}}{1 \text{ mi}} \cdot \frac{10^3 \text{ m}}{1 \text{ km}} \cdot \frac{1 \text{ ly}}{9.5 \times 10^{15} \text{ m}} \\ &= \frac{5000 \cancel{\text{mi}} \cdot 1.61 \cancel{\text{km}} \cdot 10^3 \cancel{\text{m}} \cdot 1 \text{ ly} \cdot 10^{-15}}{1 \cancel{\text{mi}} \cdot 1 \cancel{\text{km}} \cdot 9.5 \cancel{\text{m}}} = 8.47 \times 10^{-10} \text{ light years} \end{aligned}$$

5) First convert everything to inches:

$$5 \text{ feet} + 2 \text{ inches} = 5 \text{ ft} \cdot \frac{12 \text{ in}}{1 \text{ ft}} + 2 \text{ in} = 60 \text{ in} + 2 \text{ in} = 62 \text{ in}$$

Then convert the inches to meters:

$$62 \text{ in} = 62 \text{ in} \cdot \frac{2.54 \text{ cm}}{1 \text{ in}} \cdot \frac{1 \text{ m}}{100 \text{ cm}} = 1.57 \text{ m}$$

Part 4:

1)

$$30 \text{ m/s} = 30 \frac{\text{m}}{\text{s}} \cdot \frac{100 \text{ cm}}{1 \text{ m}} \cdot \frac{1 \text{ in}}{2.54 \text{ cm}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = \frac{30 \cdot 100}{2.54 \cdot 12} \frac{\text{ft}}{\text{s}} = 98.4 \text{ ft/s}$$

2)

$$600 \text{ mph} = 600 \frac{\text{mi}}{\text{hr}} \cdot \frac{1 \text{ hr}}{60 \text{ min}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 0.167 \text{ mi/s}$$

3)

$$65 \text{ mph} = 65 \frac{\text{mi}}{\text{hr}} \cdot \frac{1.61 \text{ km}}{1 \text{ mi}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot \frac{1 \text{ hr}}{60 \text{ min}} \cdot \frac{1 \text{ min}}{60 \text{ sec}} = 29.1 \text{ m/s}$$

4)

$$3 \times 10^8 \text{ m/s} = 3 \cdot 10^8 \cdot \frac{\text{m}}{\text{s}} \cdot \frac{1 \text{ km}}{10^3 \text{ m}} \cdot \frac{1 \text{ mi}}{1.61 \text{ km}} \cdot \frac{60 \text{ s}}{1 \text{ min}} \cdot \frac{60 \text{ min}}{1 \text{ hr}} = 6708 \times 10^5 \frac{\text{mi}}{\text{hr}} = 6.71 \times 10^8 \text{ mph}$$

5)

$$\begin{aligned} 3 \times 10^8 \text{ m/s} &= 3 \cdot 10^8 \frac{\text{m}}{\text{s}} \cdot \frac{1 \text{ ly}}{9.5 \cdot 10^{15} \text{ m}} \cdot \frac{60 \text{ s}}{1 \text{ min}} \cdot \frac{60 \text{ min}}{1 \text{ hr}} \cdot \frac{24 \text{ hrs}}{1 \text{ day}} = 27284 \times 10^{-7} \frac{\text{ly}}{\text{day}} \\ &= 2.73 \times 10^{-3} \frac{\text{ly}}{\text{day}} \end{aligned}$$

Part 5:

1) Area! Area has units of length squared.

2)

$$\begin{aligned} 500 \text{ ft}^2 &= 500 \text{ ft}^2 \cdot \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)^2 = \frac{500 \text{ ft} \cdot \text{ft}}{1} \cdot \frac{12 \text{ in}}{1 \text{ ft}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 72000 \text{ in} \cdot \text{in} = 72000 \text{ in}^2 \\ &= 7.2 \times 10^4 \text{ in}^2 \end{aligned}$$

3) Volume! Volume has units of length to the third power, or length cubed.

4)

$$\begin{aligned} 55 \text{ m}^3 &= 55 \text{ m}^3 \cdot \left(\frac{100 \text{ cm}}{1 \text{ m}} \right)^3 \cdot \left(\frac{1 \text{ in}}{2.54 \text{ cm}} \right)^3 = \frac{55 \text{ m}^3}{1} \cdot \frac{100^3 \text{ cm}^3}{1 \text{ m}^3} \cdot \frac{1 \text{ in}^3}{2.54^3 \cdot \text{cm}^3} = 3356306 \text{ in}^3 \\ &= 3.36 \times 10^6 \text{ in}^3 \end{aligned}$$

5)

$$\text{Area} = \text{length} \times \text{width} = 360 \text{ ft} \cdot 160 \text{ ft} = 57600 \text{ ft}^2 = 5.76 \times 10^4 \text{ ft}^2$$

Convert to cm^2 :

$$\begin{aligned} 5.76 \times 10^4 \text{ ft}^2 &= 5.76 \times 10^4 \text{ ft}^2 \cdot \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)^2 \cdot \left(\frac{2.54 \text{ cm}}{1 \text{ in}} \right)^2 = \frac{5.76 \times 10^4 \text{ ft}^2}{1} \cdot \frac{12^2 \text{ in}^2}{1 \text{ ft}^2} \cdot \frac{2.54^2 \text{ cm}^2}{1 \text{ in}^2} \\ &= 5.35 \times 10^7 \text{ cm}^2 \end{aligned}$$

Part 6:

1) In a week:

$$60 \frac{\text{beats}}{\text{min}} = 60 \frac{\text{beats}}{\text{min}} \cdot \left(\frac{60 \cdot 24 \cdot 7 \text{ min}}{1 \text{ week}} \right) = 604800 \frac{\text{beats}}{\text{week}} = 6.0 \times 10^5 \frac{\text{beats}}{\text{week}}$$

In a year:

$$604,800 \frac{\text{beats}}{\text{week}} \cdot \left(\frac{52 \text{ weeks}}{1 \text{ year}} \right) = 31449600 \frac{\text{beats}}{\text{year}} = 3.1 \times 10^7 \frac{\text{beats}}{\text{year}}$$

- 2) If we take Butler Library to be at 114th street, and that Times Square ends at 48th street, then the distance between them is 114-48=66 blocks. This corresponds to

$$66 \text{ blocks} = 66 \text{ blocks} \cdot \left(\frac{1 \text{ mi}}{20 \text{ blocks}} \right) \cdot \left(\frac{1.6 \text{ km}}{1 \text{ mi}} \right) = 5.3 \text{ km}$$

- 3) 1 gallon = 128 fluid ounces, and 3.8 liters = 1 gallon. So in 4 hours:

$$4 \text{ hours} \cdot \left(\frac{30 \text{ cups coffee}}{1 \text{ hour}} \right) \cdot \left(\frac{12 \text{ oz}}{1 \text{ cup coffee}} \right) \cdot \left(\frac{1 \text{ gal}}{128 \text{ oz}} \right) \cdot \left(\frac{3.8 \text{ liter}}{1 \text{ gal}} \right) = 42.6 \text{ liters}$$

- 4) A golden retriever eats

$$3 \frac{\text{cups}}{\text{day}} \cdot \frac{365 \text{ days}}{1 \text{ year}} \cdot \frac{100 \text{ g}}{1 \text{ cup}} \cdot \frac{1 \text{ lb}}{454 \text{ g}} \cdot \frac{\$40}{33 \text{ lb}}$$

- 5) One piano tuner can do:

$$\frac{255 \text{ days}}{\text{year}} \cdot \frac{8 \text{ hrs}}{\text{day}} \cdot \frac{1 \text{ piano}}{2 \text{ hrs}} = 1020 \frac{\text{pianos}}{\text{year}}$$

In words, this is “1020 pianos per year per piano tuner”. So we can write:

$$1020 \frac{\text{pianos}}{\text{year}} / \text{piano tuner} = 1020 \frac{\text{pianos}}{\text{year} \cdot \text{piano tuner}}$$

So, starting from the population of the city:

$$\frac{8 \times 10^6 \text{ people}}{\text{NYC}} \cdot \frac{1 \text{ piano}}{100 \text{ people}} \cdot \frac{1 \text{ piano tuner}}{1020 \frac{\text{pianos}}{\text{year}}} = 78.43 \text{ piano tuners} \rightarrow 79 \text{ piano tuners}$$



Answers to Sense of Scale Tutorial Practice Questions

Part 1:

- 1) $? = 5$
- 2) $? = 23$
- 3) $? = -4$
- 4) $? = 3$
- 5) $? = -19$
- 6) $? = -34$
- 7) $? = 6.7 \times 10^{14}$
- 8) $? = 6.7 \times 10^{-16}$
- 9) $? = 6.7 \times 10^{17}$
- 10) $? = 3.5 \times 10^{25}$

Part 2:

- 1) 10^{-15} m
- 2) 10^{-10} m
- 3) 10^{-4} m
- 4) 10^2 m
- 5) 10^8 m
- 6) 10^{11} m
- 7) 10^{16} m
- 8) 10^{20} m
- 9) 10^{27} m
- 10) 10^4 years ago

11) 10^8 years ago

12) 10^9 years

13) 10^{10} years

Note: It is fine if you don't have all of these numbers memorized, but it helps to have a few benchmarks at different scales in your head for reference

Part 3:

1) 10^{15} times bigger.

15 orders of magnitude bigger.

2) 10^{16} times bigger.

16 orders of magnitude bigger.

3) 10^6 times bigger.

6 orders of magnitude bigger.

4) 10^6 times bigger.

6 orders of magnitude bigger.

5) 10^{42} times bigger.

42 orders of magnitude bigger.

6) 3 times bigger.

0 orders of magnitude bigger.

7) 10 times bigger.

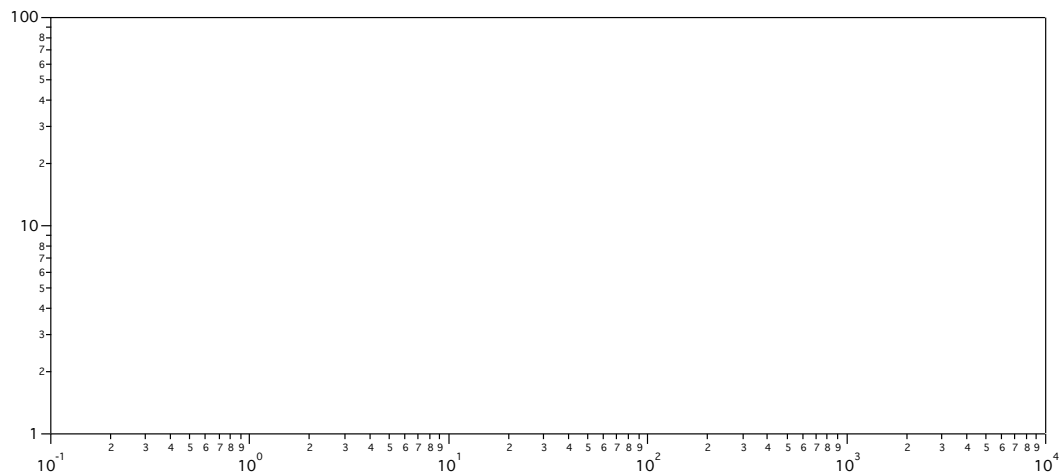
1 orders of magnitude bigger.

8) 10^4 times bigger.

4 orders of magnitude bigger.

Part 4:

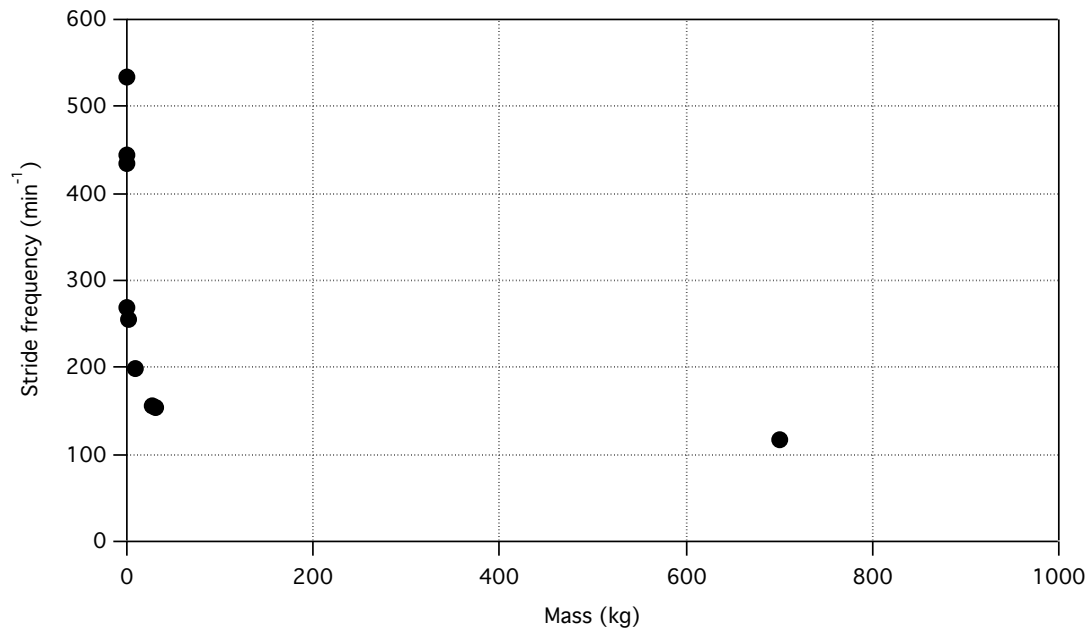
1) Fully labeled scales:



2)

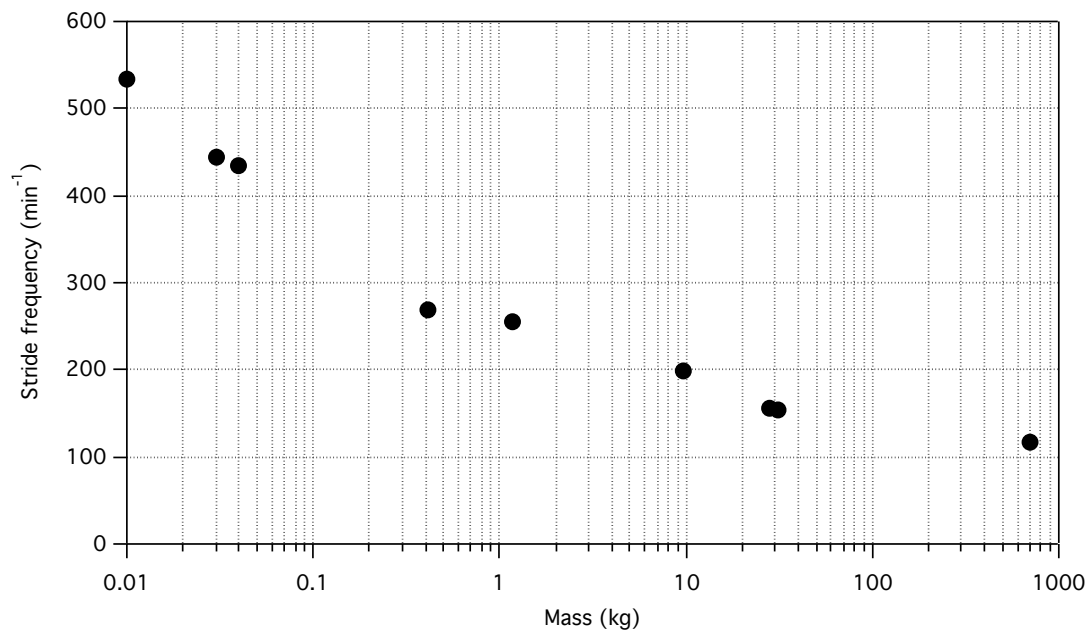
<i>Point</i>	<i>x-value</i>	<i>y-value</i>
A	0.2	2
B	0.4	25
C	3.5	18
D	4.5	7
E	20	40
F	65	12
G	150	3
H	650	10
I	1500	4
J	5000	5
K	9000	90

3)



Not very informative...

4)



Now the trend in the data is clearer!¹

¹ Since the data in this graph follows a power law, if we used a log-log plot instead of the semi-log plot shown here, the points would lie on a straight line, and the trend would be even better elucidated.



Answers to Probability Tutorial Practice Questions

Probability of mutually exclusive events:

- 1) $P(\text{July } 4^{\text{th}}) = 1/365 = 0.00274 = 0.274\%$ (ignoring leap day birthdays)
- 2) $P(\text{July } 4^{\text{th}} \text{ OR } \text{Feb } 14^{\text{th}}) = P(\text{July } 4^{\text{th}}) + P(\text{Feb } 14^{\text{th}}) = \frac{1}{365} + \frac{1}{365} = \frac{2}{365} = 0.00548 = 0.548\%$
- 3) Zero, of course, since the two events (a July 4th birthday and a Feb 14th birthday) are mutually exclusive. Conversely, if the probability of A and B both occurring is zero, you can conclude A and B are mutually exclusive.

Probability of independent events

- 4) $1/365 = 0.00274 = 0.274\%$
- 5) $(1/365) \cdot (1/2) = 0.137\%$
- 6) $(1/365) \cdot (1/365) = 7.51 \times 10^{-6}$

Probability of rare events

- 7) Assuming there are 20 other people in your seminar,

$$P = 1 - (1 - P(A))^N = 1 - \left(1 - \frac{1}{365}\right)^{20} = 0.0534 = 5.34\%$$

Note that since this is a small probability, it is well approximated by

$$P = N \cdot P(A) = 0.0548 = 5.48\%$$

- 8) Assuming there are 500 other people currently taking Frontiers of Science,

$$P = 1 - (1 - P(A))^N = 1 - \left(1 - \frac{1}{365}\right)^{500} = 0.746 = 74.6\%$$

Note that since this probability is not small, $N \cdot P(A)$ is a poor approximation. In fact, it gives a non-sensical answer:

$$P = N \cdot P(A) = 1.37 = 137\% \rightarrow \text{nonsense!}$$

- 9) Approximating the probability of choosing a number between 1 and 50 is $1/50$,

$$P = 1 - (1 - P(A))^N = 1 - \left(1 - \frac{1}{50}\right)^{20} = 33.2\%$$

Again, this probability is not small, so the approximation doesn't work:

$$P = N \cdot P(A) = 40\% \text{ (substantially different from 33.2\%)}$$

Note that the approximation breaks down when either N or $P(A)$ get large enough to make P too large. The best way to check if the approximation works is to calculate the probability using the true formula and compare your answer to the approximation.

Probability of rare events from past occurrence rates

- 10) If we assume that 2011 was a typical year for lightning strikes, then we can take the occurrence rate for this year to be the probability of being struck by lightning in one year. In 2011 there were 400 lightning strikes for 310×10^6 people, so the occurrence rate per person for that year was:

$$O_{\text{year}} = \frac{400}{310 \times 10^6} = 1.3 \times 10^{-6} / \text{year}$$

- 11) If we assume that the lightning strikes were distributed evenly throughout the year, so that the occurrence rate of lightning strikes was about the same for each day of the year, we can divide the rate per year by 365 to get the rate per day. We can also think of this as a unit conversion from years to days:

$$O_{\text{day}} = \frac{1.3 \times 10^{-6}}{\text{year}} \cdot \frac{1 \text{ year}}{365 \text{ days}} = 3.5 \times 10^{-9} / \text{day}$$

- 12) Since the number of opportunities in this question is 10 days, an "opportunity" is a day. Set $N = 10$ and use the occurrence rate per day as the probability given one opportunity $P(A) = O_{\text{day}}$

$$P = 1 - (1 - P(A))^N = 1 - (1 - 3.5 \times 10^{-9})^{10} = 3.5 \times 10^{-8}$$

- 13) Since the number of opportunities in this question is 80 years, we set $N = 80$, and use the occurrence rate per year as the probability $P(A) = O_{\text{year}}$

$$P = 1 - (1 - P(A))^N = 1 - (1 - 1.3 \times 10^{-6})^{80} = 1.0 \times 10^{-4}$$